SUBRAMANI RAJA R

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Portfolio: https://subramani-raja.vercel.app

OBJECTIVE

Problem Solving **Full Stack Developer** strong in Algorithms, Data Structures, Object-Oriented Programming, and full-stack web development (React, Node.js, MySQL/MongoDB). Proven ability to design and deliver **scalable**, optimized, **maintainable**, high-performance solutions. Backed by a published research project and a role-based full-stack application. Seeking an entry-level Software Developer / Full Stack Developer role.

TECHNICAL SKILLS

Programming Languages: Java, C, Python

Web Development: HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, PHP, JWT

Databases: MySQL, MongoDB

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Software Development Life Cycle (SDLC), REST APIs

Tools & Platforms: Git, GitHub, Postman, Figma, Chrome Dev Tools

Familiar With: TensorFlow, Unity3D, Vuforia, 3D Modelling, Blockchain Technologies

EDUCATION

B.E., Computer Science and Engineering (Completed) 2022 – 2026
Ramco Institute of Technology | CGPA: 8.13

HSC — Raja Matric Hr Sec School 2022
Percentage: 78%

PROJECTS

Mobile Shop Management System — MERN Stack Application *2025*

- Developed a full-stack mobile shop management system using the MERN stack (MongoDB, Express.js, React.js, Node.js) to manage inventory, sales, billing, and customer records.
- Built RESTful APIs for product catalog, stock, and billing operations, with role-based access for admin and staff users.
- Designed a responsive React.js dashboard for real-time sales tracking, stock alerts, and reporting.
- Modeled MongoDB schemas for products, transactions, and customers to support efficient queries.

Tech Stack: *React.js, Node.js, Express.js, MongoDB, JavaScript*

Inventory Management System — Full-Stack Web Application *2024*

- Designed and developed a role-based full-stack system (Admin, Manager, Customer) with secure, controlled access for each user type.
- Built an end-to-end order processing workflow with CRUD operations and automated stock-level updates on dispatch.
- Designed a normalized MySQL relational database schema for efficient inventory storage and retrieval.

Tech Stack: *PHP, MySQL, HTML, CSS, JavaScript*

AI-Powered Real-Time Safety Monitoring System — Conference Publication *Present – 2026*

- Architected a modular three-stage software pipeline (identification, pose estimation, behavior analysis), achieving 98.4% accuracy at ~15 FPS in real time.
- Built a scalable identification module using ArcFace embeddings, allowing new users to be added without retraining the model.
- Co-authored and published findings in a peer-reviewed conference paper, demonstrating research, documentation, and technical communication skills.

Tech Stack: *Python, OpenCV, RetinaFace, ArcFace, YOLOv7-Pose, Bi-LSTM, TensorFlow/PyTorch, ONNX Runtime*

Audio Query Information Retrieval System — ASR & NLP Backend *2024*

- Built a voice-based query-retrieval backend for farmers using Whisper ASR for speech-to-text and SBERT embeddings with FAISS for fast semantic search.
- Improved retrieval accuracy and system efficiency through embedding fine-tuning, metadata filtering, and compression; added Kannada-language support for accessibility.

Tech Stack: *Python, Whisper ASR, SBERT, FAISS*

CERTIFICATIONS & ACHIEVEMENTS

- AWS Academy Graduate — Cloud Architecting (Training Badge)
- **Java Fundamentals — Wipro TalentNext (Certified Java Developer)** | Programming in Java — NPTEL
- Introduction To Front End Design | Foundation Of Ux Design | Build Wireframes And Low Fidelity Prototypes – Coursera
- Participant, ML-Fiesta: AI/ML Hackathon, Synergy'24, IIT Bangalore